Colin Morris-Moncada
D06279659

Homework 2

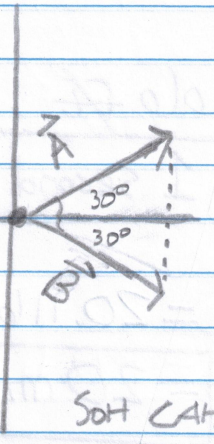
1. The figure shows a traction apparatus which is used to prevent muscle contraction on a patients leg. The pull of the two cords, which are presented by the vectors $\vec{A}$ and $\vec{B}$, have equal magnitude and combine to produce an outward horizontal traction force of 10N on the patients leg.

Find the values of each pull ($|\vec{A}|$ and $|\vec{B}|$)

$F_{traction} = 10N$
Angle $= 30°$

$\vec{A} = 5.77N$
$\vec{B} = 5.77N$

$F_{Ax} = 5N$
$F_{Bx} = 5N$
$\rightarrow F_{Ax} + F_{Bx} = F_{traction}$
$5+5 = 10$
$10 = 10$ ✓



$F_{traction} = \vec{A}\cos(30°) + \vec{B}\cos(30°)$

SOH CAH TOA

Check my work → $\vec{A}\cos(30°) + \vec{B}\cos(30°) = 10$
$5.77\cos(30°) + 5.77\cos(30°) = 10$
$10 = 10$ ✓

$F_{Ax} = F_A\cos(30°)$ $\quad$ $F_{Bx} = F_B\cos(30°)$

$F_A = \frac{F_{Ax}}{\cos(30°)}$ $\quad$ $F_B = \frac{F_{Bx}}{\cos(30°)}$

$= \frac{5}{\cos(30°)}$ $\quad$ $= \frac{5}{\cos(30°)}$

$= 5.77$ Newtons $\quad$ $= 5.77$ Newtons

2. Convert the following to basic SI units or a combination of basic SI units (kilograms, meters, and seconds - kg, m, s)

a. 8.0 in → 2 sig figs
b. 66 ft/s → 2 sig figs
c. 60 mph → 1 sig figs
d. 14 in^2 → 2 sig figs

a. $8.0\,\text{in} \cdot \frac{0.0254\,\text{m}}{1\,\text{in}} = 0.2032\,\text{m}$ $\quad 1\,\text{in} = 0.0254\,\text{m}$

$= 0.20\,\text{m}$

b. $\frac{66\,\text{ft}}{1\,\text{second}} \cdot \frac{0.3048\,\text{m}}{1\,\text{ft}}$ $\quad 1\,\text{ft} = 0.3048\,\text{m}$

$= 20.1168$ $\quad 1\,\text{mi} = 1{,}609.34\,\text{m}$

$= 20\,\text{m/s}$

c. $\frac{60\,\text{miles}}{1\,\text{hour}} \cdot \frac{1{,}609.34\,\text{m}}{1\,\text{mile}} \cdot \frac{1\,\text{hour}}{60\,\text{min}} \cdot \frac{1\,\text{min}}{60\,\text{s}}$

$= 26.8\,\text{m/s}$

d. $14\,in^2 \cdot \left(\frac{0.0254\,m}{1\,in}\right)^2 = 14 \cdot 6.4516 \times 10^{-4}\,m^2$

$= 0.00903224\,m^2$

$1\,in = 0.0254\,m$ $\quad \boxed{= 9.0 \times 10^{-3}\,m^2}$

* Remember to always square conversion factor for in^2 (square inches)

3. A runner on a straight track covers 1km in 4 min. What is their average velocity in

$\downarrow$

$\frac{1\,km}{4\,min}$

a. km/min
b. km/sec
c. km/h

a. km/min

$\frac{1\,km}{4\,min} = \boxed{0.25\,km/min}$

b. km/sec

$\frac{0.25\,km}{1\,min} \cdot \frac{1\,min}{60\,seconds}$

$= 0.00417\,km/sec$

$\boxed{= 4.17 \times 10^{-3}\,km/sec}$

c. km/h

$$\frac{0.25\ \text{km}}{1\ \text{min}} \cdot \frac{60\ \text{min}}{1\ \text{hour}}$$

$= 15\ \text{km/hour}$

4. Perform the following calculations with the correct number of significant figures

a. 159.31×204.6 → 4 sig figs
b. $5.1125 + 0.67 + 3.2$ → 2 sig figs
c. $7.662 - 7.425$ → 4 sig figs
d. $16.5 / 3.45$ → 3 sig figs

a. $159.31 \times 204.6 = 32,594.826$

$= 3.260 \times 10^4$

b. $5.1125 + 0.67 + 3.2 = 8.9825$

$= 9.0$

c. $7.662 - 7.425 = 0.237$

d. $16.5 / 3.45 = 4.7826$

$= 4.78$

5. The quantity called mass density is the mass per unit volume of a substance What are the mass densities basic SI units (kilograms, meters, and seconds) of the following objects

$1cm^3 = 1 \times 10^{-6} m^3$

a. 215 cm^3 solid with a mass of 0.0179 kg
b. 95 cm^3 liquid with a mass of 77g

a. 215 cm^3

$\rho = \frac{m}{V}$

Mass = 0.0179 kg

$215 cm^3 = 215 \times 10^{-6} m^3$
$= 2.15 \times 10^{-4} m^3$

$\rho = \frac{0.0179 kg}{2.15 \times 10^{-4} m^3}$

$= 83.3 kg/m^3$

b. 95 cm^3

Mass = 77g

$95 cm^3 = 95 \times 10^{-6} m^3$
$= 9.5 \times 10^{-5} m^3$

$\rho = \frac{0.077 kg}{9.5 \times 10^{-5} m^3}$

$= 810 kg/m^3$

6. Is the particle in the figure speeding up? Slowing down? or can you tell? Explain

The particle is slowing down because the distance between each position is decreasing as you go from left to right.

7. Does the object represented in the figure above have a positive or negative value of a_x? Explain

Object is moving in the negative x direction (leftward) and the dots are getting closer, telling us the object is slowing down

For an object to slow down the acceleration must point opposite to the velocity, so in the positive x direction

Therefore, a_x is positive

8. Does the object represented in the figure above have positive or negative a_y? Explain

The object is downward (negative y direction) but is slowing down where the distance between each dot is decreasing. Therefore, a_y is positive

9. Determine the signs (positive, negative, or zero) of the position, velocity, and acceleration for the particle in the figure above.

position = negative
velocity = positive
acceleration = positive

→ dots are below ∅
→ arrows point upward

↳ dots are getting farther apart (speeding up)

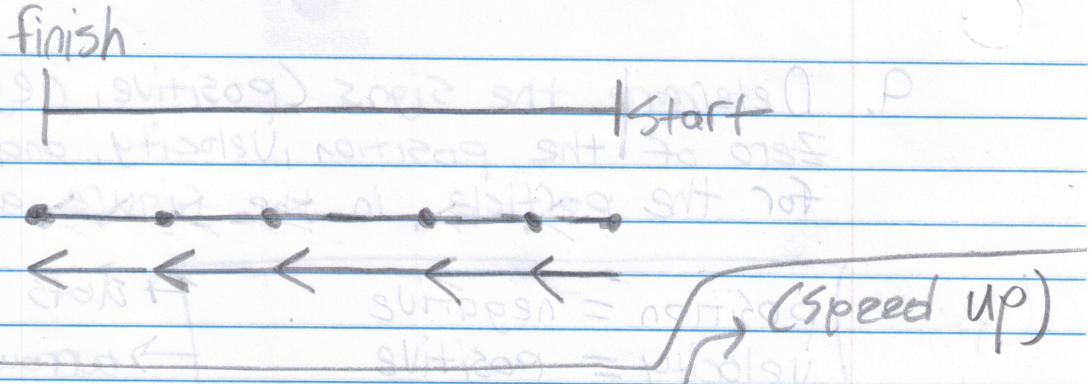
10. Determine the signs (positive, negative, or zero) of the position, velocity, and acceleration for the particle in the figure above

position = positive
velocity = positive
acceleration = positive

→ dots are above ∅
→ arrows point upward

↳ dots are getting further apart (speeding up)

11. A baseball player stats running to the left to catch the ball as soon as the hit is made. Use the particle model to draw a motion diagram showing the position and average velocity vectors of the player during the first few seconds of the run.

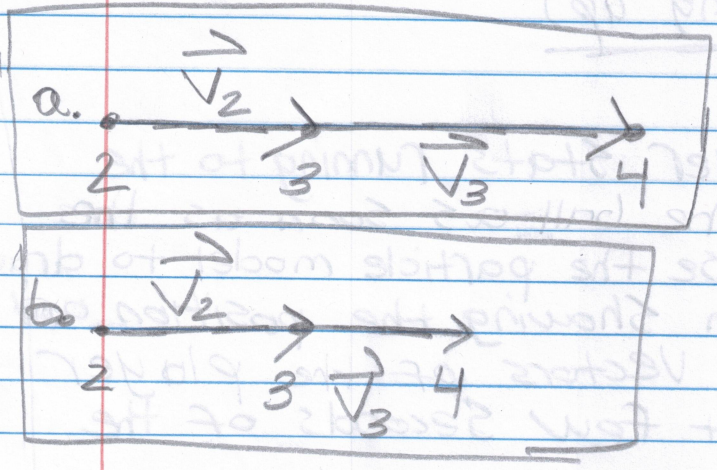


Leftward velocity arrows increase, peak, then decrease (slow down) to catch the ball

12. The figure above shows two dots of a motion diagram and vector $\vec{v}_2$
Copy this figure, then add dot 4 and the next velocity $\vec{v}_3$

a. if the acceleration vector $\vec{a}$ at dot 3 points right

b. if the acceleration vector $\vec{a}$ at dot 3 points left.



13. Write a short description of the motion of a real object for which the figure above would be a realistic position versus time graph

Runner starts a sprint 9 meters from finish line.

Runner walks toward the line moving faster in first 10 seconds then slowing down for the remainder 20 seconds

14. A car travels in the positive x-direction for 20 km at 40 km/h. It then continues in the same direction for another 20 km at 80 km/h

a. What is the average velocity of the car during this 40 km trip?

b. What is the average speed?

a. avg velocity = $\frac{\text{total displacement}}{\text{total time}}$

First leg: time = $\frac{20}{40} = \frac{1}{2} = 0.5h$

Second leg: time = $\frac{20}{80} = \frac{1}{4} = 0.25h$

Total time = $0.75h$

$$V_{avg} = \frac{40\,km}{0.75h}$$

$$= 53\,km/h$$

b. total distance = total displacement

$$= 40\,km$$

$$S_{avg} = \frac{40\,km}{0.75h}$$

$$= 53\,km/h$$

Motion is entirely in one direction so part a and part b are equal

15. Suppose the motion of the particle in the above figure is described by the equation $x = a + bt^2$, where $a = 10m$, $b = 2m/s^2$

a. Find the displacement of the particle in the time interval between $t_i = 2s$ and $t_f = 4s$

b. Find the average velocity and the average speed during this interval of time.

a. $X(t_i) = 10 + 2(2)^2$

$= 10 + 8$

$= 18\,m$

$X(t_f) = 10 + 2(4)^2$

$= 10 + 32$

$= 42\,m$

Displacement $= X(t_f) - X(t_i)$

$= 42 - 18$

$= 24\,m$

b. $\Delta t = 4 - 2$

$= 2\,s$

$V_{avg} = \frac{\Delta x}{\Delta t}$

$= \frac{24\,m}{2\,s}$

$= 12\,m/s$ (positive X-direction)

$S_{avg} = \frac{24\,m}{2\,s}$

$= 12\,m/s$

16. A graph of position (in meters) versus time (in seconds) for a person traveling in the positive x-direction is displayed in the figure above. Find the average velocity for the following cases

a. $t_i = 2s$ and $t_f = 4s$

$$V_{avg} = \frac{(16-2)}{(4-2)}$$

$$= \frac{14}{2}$$

$$= 7 m/s$$

b. $t_i = 2s$ and $t_f = 6s$

$$V_{avg} = \frac{(38-2)}{(6-2)}$$

$$= \frac{36}{4}$$

$$= 9 m/s$$

c. $X_i = 12m$ and $X_f = 30m$

$$V_{avg} = \frac{(30-12)}{(5-2.5)} = \frac{18}{2.5} = 7.2 m/s$$

17. A body moves along a straight line with position given by $X = 8t - 2t^2$ where X is in meters and t is in seconds. Find the average velocity and average speed of the body in the intervals

a. From $t_i = 0$ to $t_f = 2s$

$$X = 8t - 2t^2$$

$$X(0) = 8(0) - 2(0)^2$$
$$= 0m$$

$$X(2) = 8(2) - 2(2)^2$$
$$= 16 - 8$$
$$= 8m$$

$$V_{avg} = \frac{(8-0)}{(2-0)} = 4m/s$$

$$V = \frac{dX}{dt} = 8 - 4t = 0$$

$$t = 2s$$

$$S_{avg} = 4m/s$$

b. $t_i = 0$ to $t_f = 5s$

$x(0) = 0m$

$x(5) = 8(5) - 2(5)^2$

$= 40 - 50$

$= -10m$

$v_{avg} = \frac{(-10 - 0)}{(5 - 0)} = -2 m/s$

$x(2) = 8m$

total distance (displacement) $= 8 + 18$

$= 26m$

$s_{avg} = \frac{26}{5} = 5.2 m/s$